

Apriori Algorithm: Detailed Lecture Notes with Fully Solved Numerical Examples and Graphical Illustrations

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1 Introduction

The Apriori algorithm is widely used for **frequent itemset mining** and **association rule learning**. It identifies patterns or associations among items in datasets such as market-basket data or document word co-occurrences.

2 Why Apriori is Used

- Discover frequently co-occurring items in transactions.
- Build association rules (“If A then B”) for recommendations.
- Reduce search space using the Apriori principle.
- Support analytics in retail, text mining, and web logs.

3 Key Definitions

Itemset: A set of items (e.g., {A,B}).

Support: Fraction or count of transactions containing the itemset.

$$\text{support}(X) = \frac{|\{t \in T : X \subseteq t\}|}{|T|}$$

Minimum Support (minsup): The threshold below which an itemset is considered infrequent and eliminated. **Confidence:**

$$\text{confidence}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)}$$

Lift:

$$\text{lift}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A) \cdot \text{support}(B)}$$

4 Apriori Principle

If an itemset is frequent, all its subsets must be frequent. Candidates with any subset below minsup are eliminated immediately.

5 Apriori Pseudocode

1. Scan DB to get C1 (counts of individual items).
2. L1 = items in C1 with support $\geq$ minsup.
3. For $k = 2$ while L_{k-1} not empty:
 - a. Generate C_k by joining L_{k-1} with itself.

- b. Prune C_k : remove candidates with any subset not in L_{k-1} .
 - c. Scan DB to count supports of C_k .
 - d. Eliminate candidates with support $< \text{minsup}$.
 - e. L_k = remaining candidates.
- 4. Return all frequent itemsets = union of all L_k .
 - 5. Generate association rules and evaluate confidence/lift.

6 Fully Worked Example Market Basket

Dataset:

TID	Items
<i>T1</i>	<i>A, B, C</i>
<i>T2</i>	<i>A, C</i>
<i>T3</i>	<i>B, C</i>
<i>T4</i>	<i>A, B</i>
<i>T5</i>	<i>A, B, C</i>

Total transactions: 5, minimum support = 40% $\Rightarrow$ count threshold = 2.

6.1 Step 1 Candidate 1-itemsets (C1)

Item	T1	T2	T3	T4	T5	Count	Meets minsup?
A	1	1	0	1	1	4	Yes
B	1	0	1	1	1	4	Yes
C	1	1	1	0	1	4	Yes
D	0	0	0	0	0	0	No

Table 1: C1 membership and elimination based on minimum support.

$$L1 = \{A, B, C\}$$

6.2 Step 2 Candidate 2-itemsets (C2)

Generate all 2-combinations from L1: AB, AC, BC.

Membership and elimination:

Itemset	T1	T2	T3	T4	T5	Count	Meets minsup?
AB	1	0	0	1	1	3	Yes
AC	1	1	0	0	1	3	Yes
BC	1	0	1	0	1	3	Yes
BD	0	0	0	0	0	0	No

Table 2: C2 membership and pruning based on minsup.

$$L2 = \{AB, AC, BC\}$$

6.3 Step 3 Candidate 3-itemsets (C3)

Join L2 to form ABC. Only candidate:

Itemset	T1	T2	T3	T4	T5	Count	Meets minsup?
ABC	1	0	0	0	1	2	Yes
ABD	0	0	0	0	0	0	No

Table 3: C3 membership and elimination based on minsup.

$$L3 = \{ABC\}$$

6.4 Summary of Frequent Itemsets

Itemset	Count	Support
A	4	0.8
B	4	0.8
C	4	0.8
AB	3	0.6
AC	3	0.6
BC	3	0.6
ABC	2	0.4

6.5 Association Rules Example

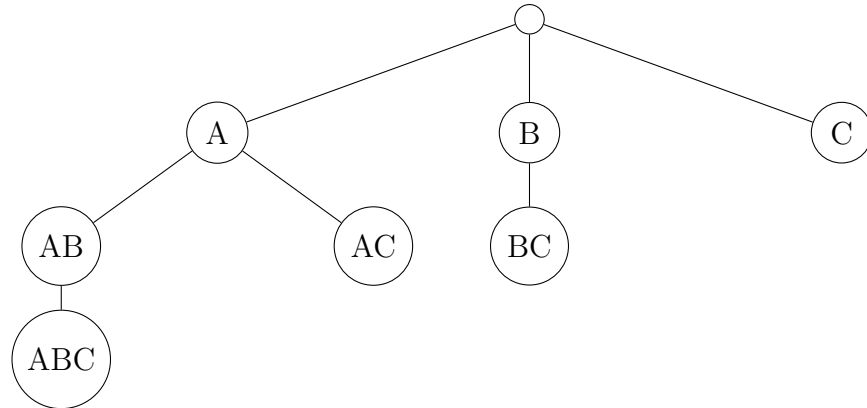
$$\text{Confidence}(A \rightarrow B) = \frac{3/5}{4/5} = 0.75$$

$$\text{Lift}(A \rightarrow B) = \frac{3/5}{(4/5)(4/5)} = 0.9375$$

$$\text{Confidence}(AB \rightarrow C) = \frac{2/5}{3/5} = 0.6667$$

$$\text{Lift}(AB \rightarrow C) = \frac{2/5}{(3/5)(4/5)} = 0.8333$$

7 Graphical Illustration Frequent Itemset Tree



Explanation: - Nodes = itemsets. - Paths from root = candidate combinations. - Only nodes meeting minsup are retained (greyed/eliminated nodes are discarded).

8 Practical Example: Google Docs Word Co-occurrence

Treat each document as a transaction and each keyword as an item:

DocID	Words
<i>D1</i>	<i>project, meeting, review</i>
<i>D2</i>	<i>project, deadline, budget</i>
<i>D3</i>	<i>meeting, review, project</i>
<i>D4</i>	<i>deadline, budget</i>
<i>D5</i>	<i>project, budget, meeting</i>

8.1 Step 1 C1 Counts

Word	Count	Support
project	4	0.8
meeting	3	0.6
review	2	0.4
deadline	2	0.4
budget	3	0.6

8.2 Step 2 C2 Counts (Pairs)

Word Pair	Count	Support
project & meeting	3	0.6
project & budget	2	0.4
meeting & review	2	0.4
deadline & budget	2	0.4

8.3 Association Rules

$$\text{Confidence}(\textit{project} \rightarrow \textit{meeting}) = 0.75$$

$$\text{Lift}(\textit{project} \rightarrow \textit{meeting}) = 0.625$$

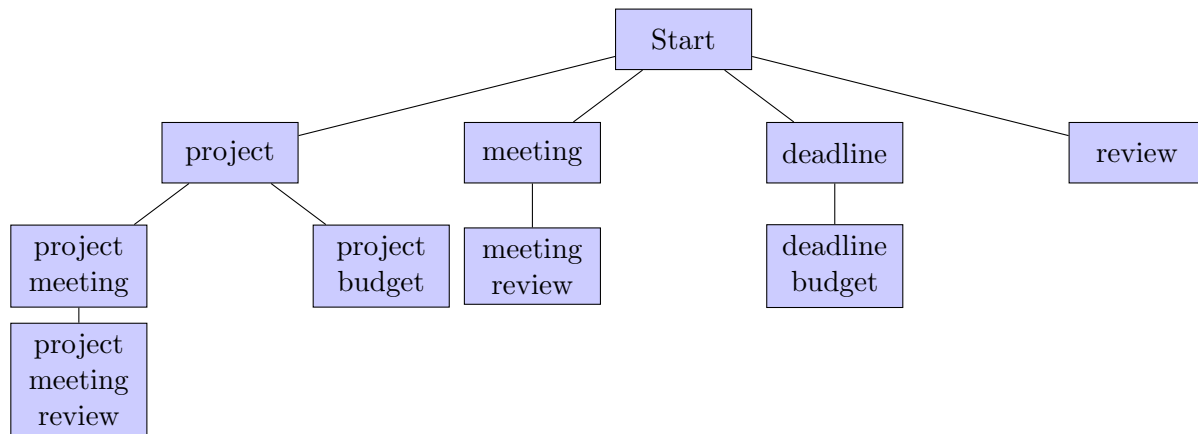
$$\text{Confidence}(\textit{deadline} \rightarrow \textit{budget}) = 1.0$$

$$\text{Lift}(\textit{deadline} \rightarrow \textit{budget}) = 0.6667$$

8.4 Interpretation

- “project & meeting” can guide autocomplete.
- “deadline & budget” can suggest financial sections in templates.

8.5 Frequent Itemset Tree Google Docs



9 Advantages and Disadvantages

9.1 Advantages

- Simple and easy to understand.
- Candidate pruning reduces computation.
- Works well for small-medium sparse datasets.

9.2 Disadvantages

- Multiple passes over database required.
- Candidate explosion for dense datasets.
- Not suitable for very high-dimensional datasets.
- Interpreting too many rules requires thresholds.